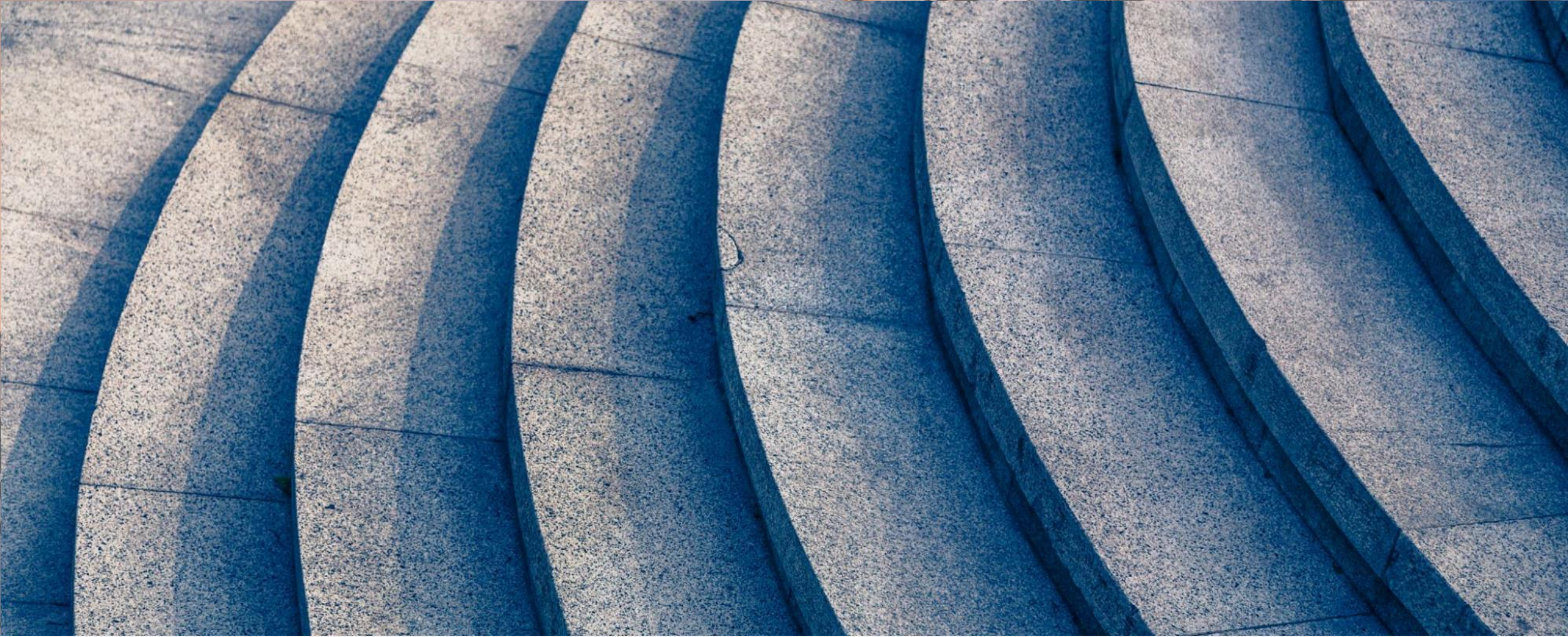




Comparative Analysis of Machine Learning and Deep Learning Tools and Frameworks.

Presented by Melvis Maduagwu

Presentation Outline

What is TensorFlow?

- What are Tensors
- TensorFlow Attributes
- TensorFlow special features
- TensorFlow Ecosystem

What is PyTorch?

- PyTorch attributes
- PyTorch special features
- PyTorch Ecosystem

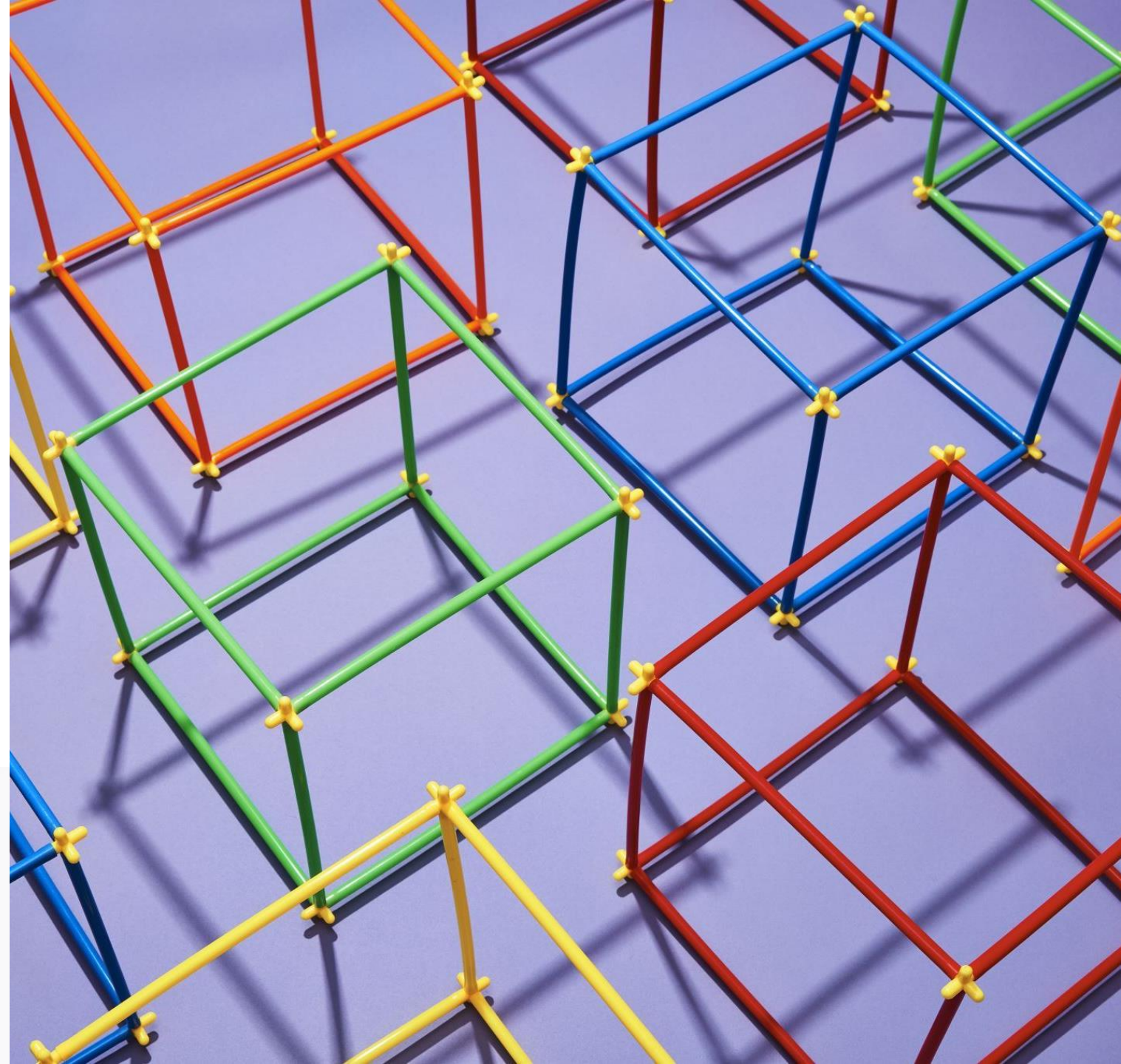
PyTorch vs TensorFlow Decision Guide

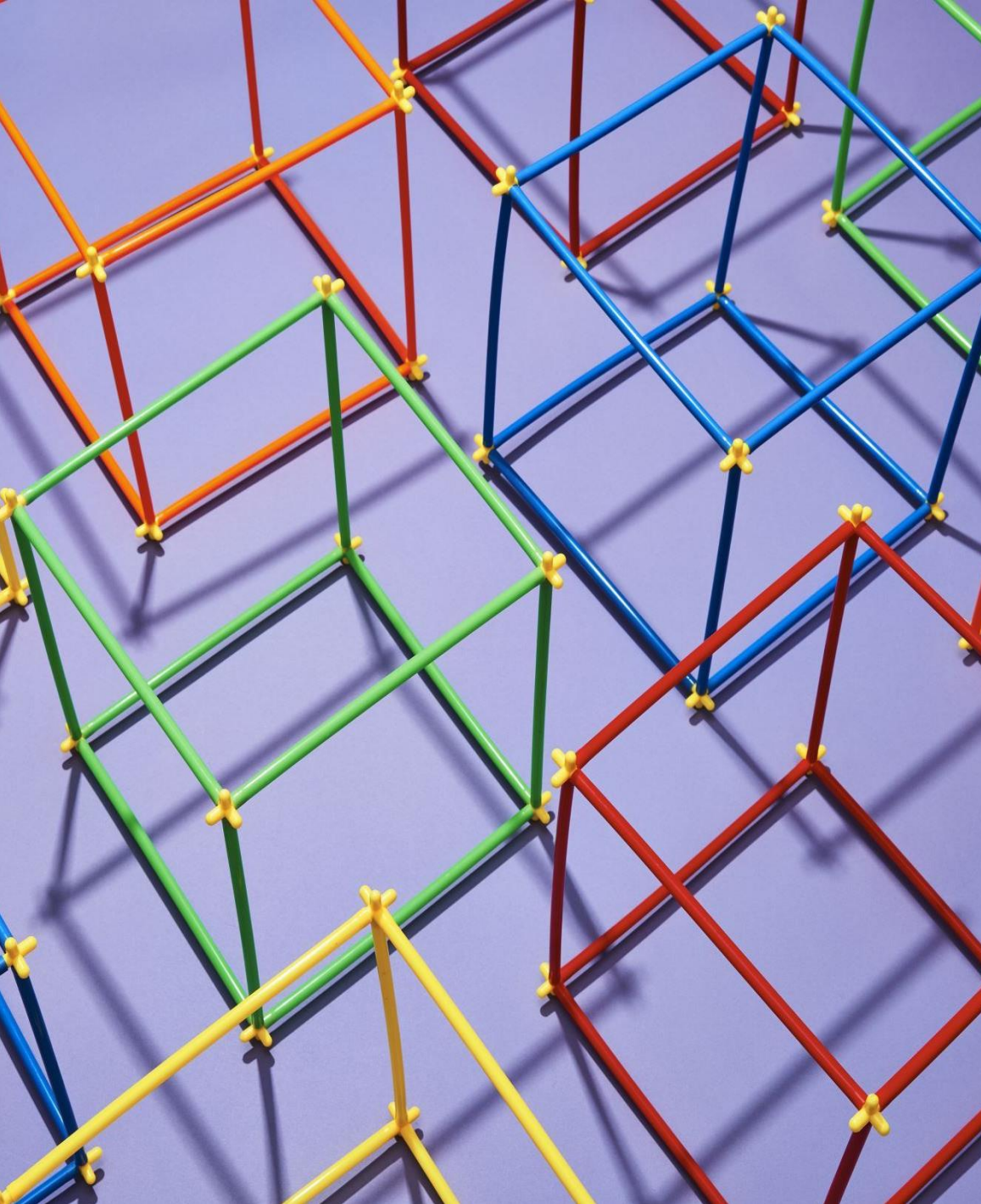
- Style
- Model
- Datatypes

Conclusion

What is TensorFlow?

TensorFlow is an open-source end-to-end platform developed by Google for building Machine Learning and Deep Learning models. It supports multi-dimensional array numeric computation, Automatic differentiation, GPU and distribution systems processing, facilitating model construction and export.





What are Tensors?

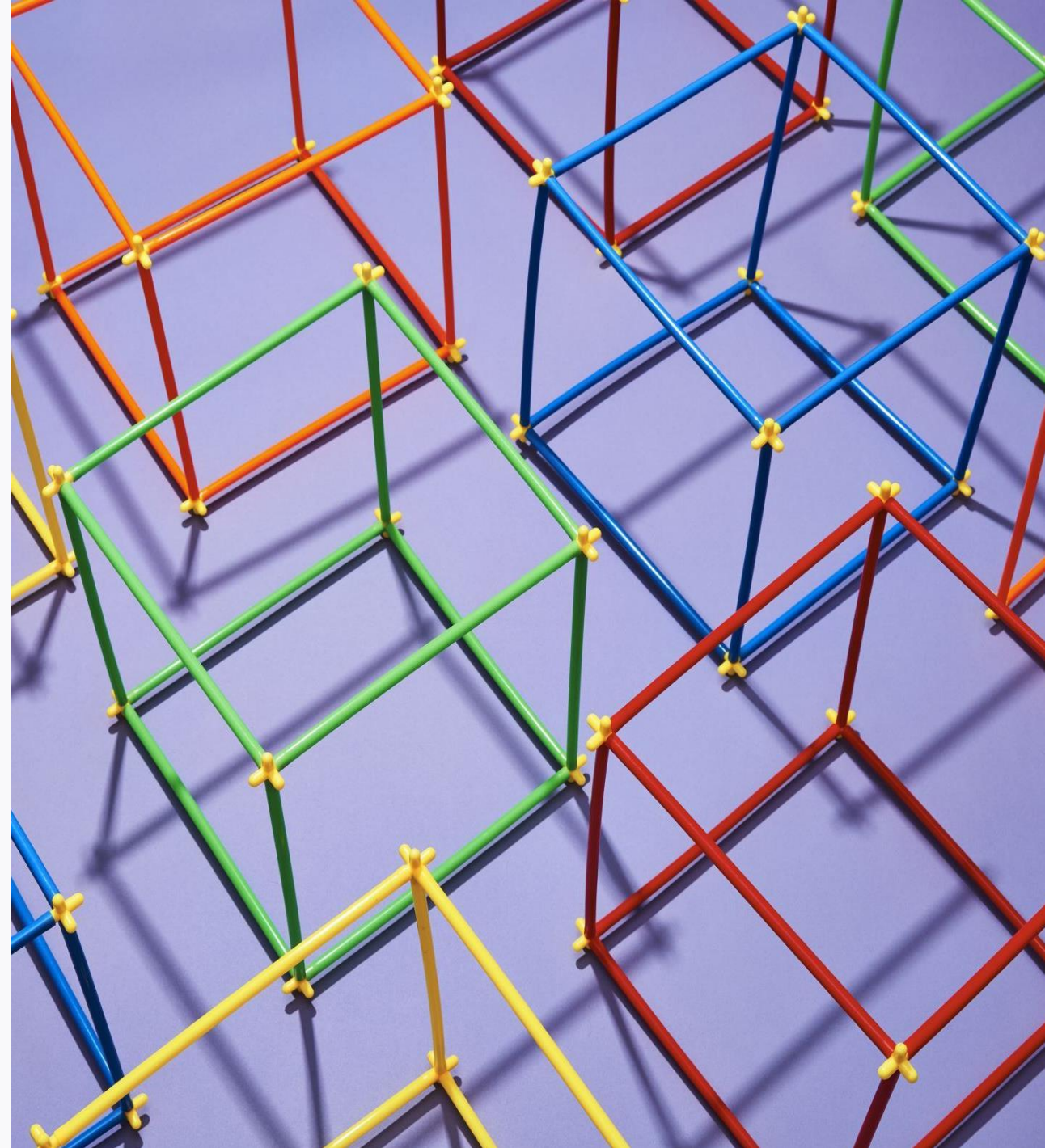
Tensors are multi-dimensional arrays with a uniform type called (dtype). We can see all the supported dtypes at `tf.dtypes`.

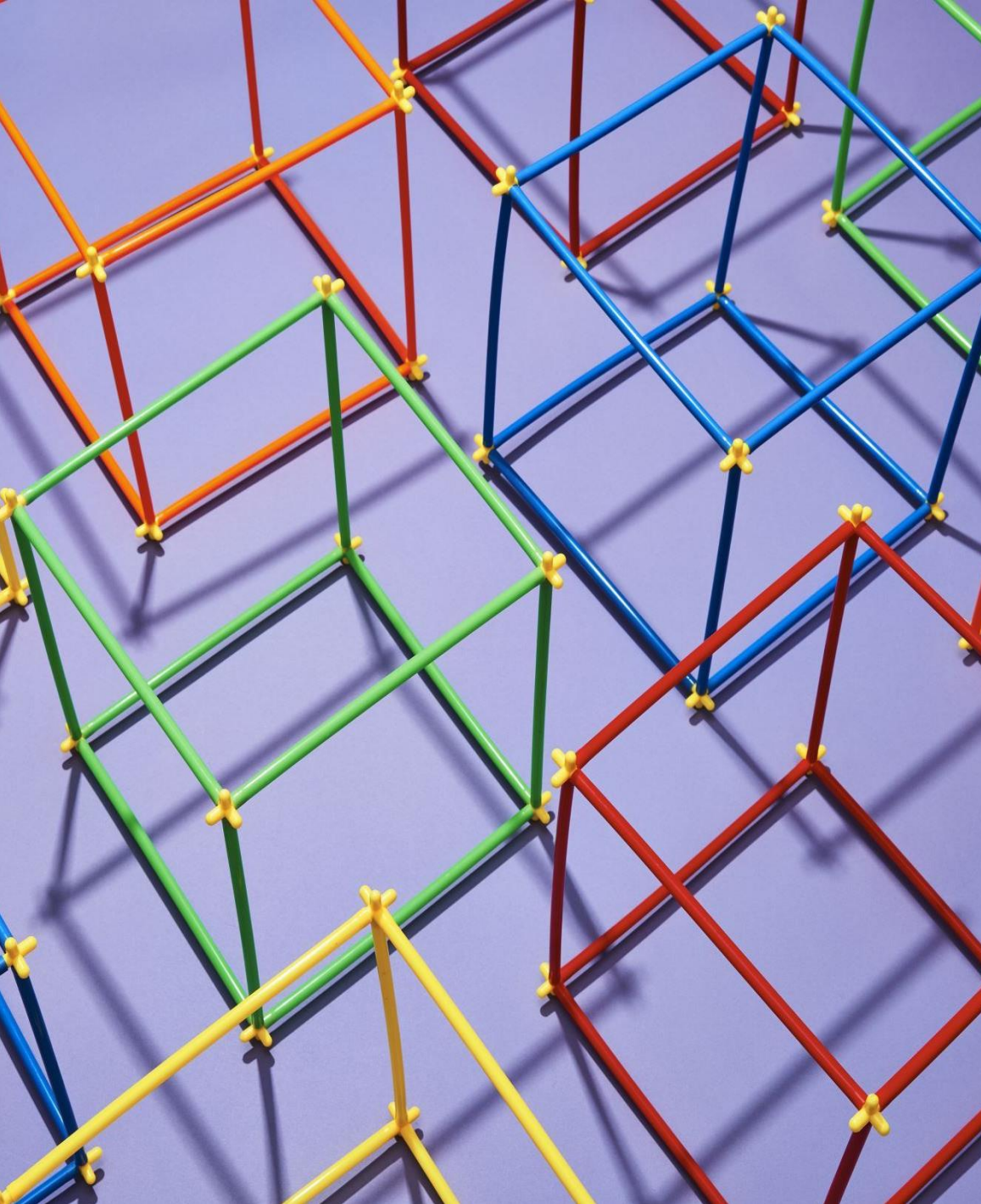
They are like the `np.arrays` in NumPy. Tensors are immutable just like Python strings and numbers, meaning we cannot change the contents of an existing tensor but only create a new one. Tensors also have different levels:

- Scalar or Rank-0 tensor: A scalar contains a single value and no axes
- Vector or Rank-1 tensor: This is like a list of values. A vector has one axis.
- Matrix or Rank-2 tensor: This has two axes
- And then multi-dimensional with three or more axis.

TensorFlow Attributes

- Shape: A tuple of integers that define the size of the tensor along each axis or dimension. For example, a 2D tensor (matrix) with 3 rows and 4 columns has a shape of (3,4).
- Data type (dtype): This specifies the data type of the elements stored within a tensor such as a 32-bit floating point (float32), a 64-bit integer (int64) or a boolean (bool).



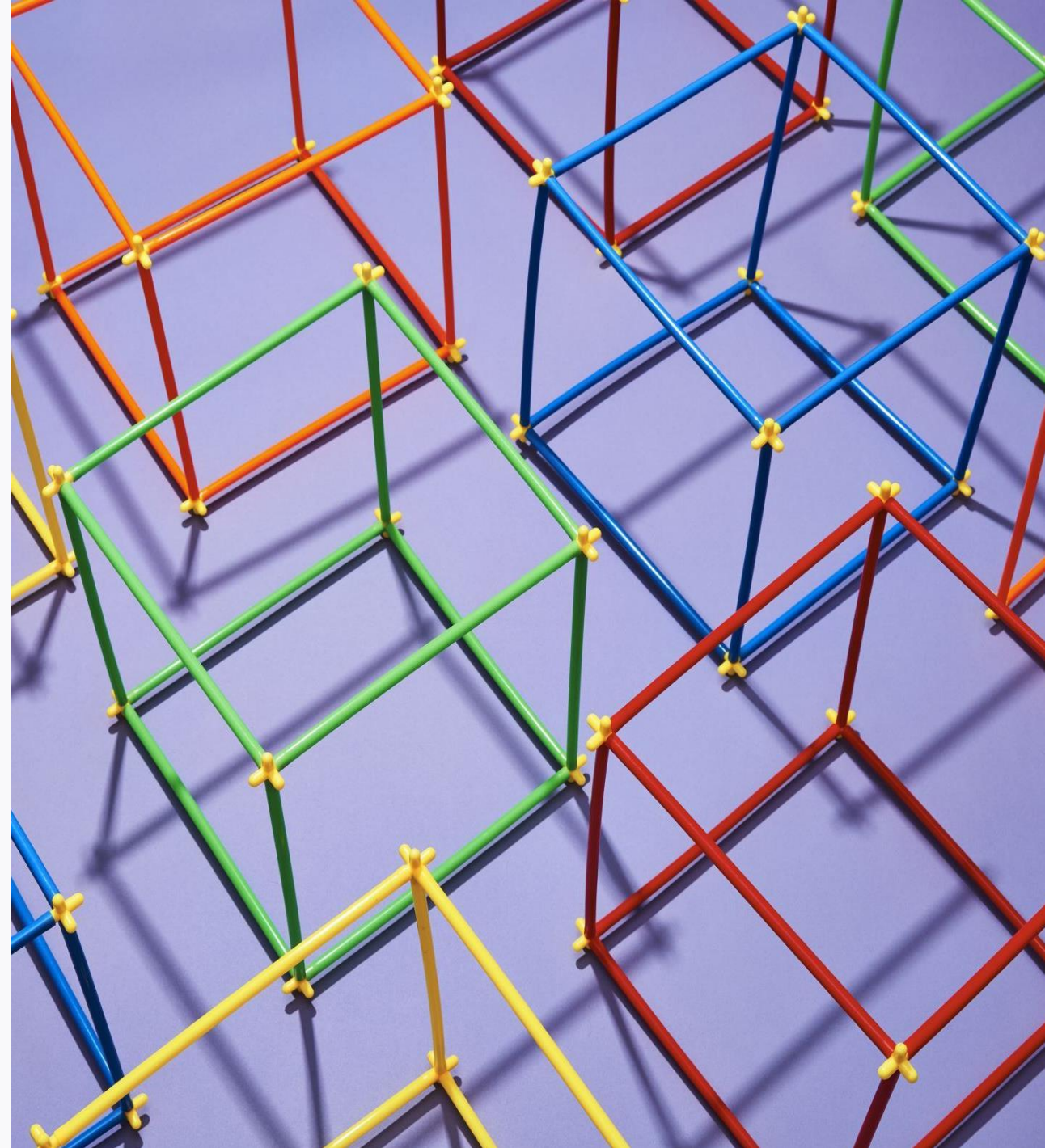


TensorFlow ecosystem

- TensorFlow core: Main library for building and training deep learning models.
- Keras: High-level API for fast, user-friendly model development.
- TensorFlow Datasets(TFDS): Ready-to-use training datasets for training and evaluation.
- TensorFlow Hub: Pretrained models and reusable ML components.
- TensorFlow Lite: Deploy models on mobile and embedded devices.
- TensorFlow.js: Run and train models in the browser or with JavaScript.
- TensorBoard: Visualization and debugging tool for models and training

TensorFlow special features

- Production and deployment strength: Easy deployment to mobile , web and embedded devices.
 - Graph execution (static graphs): `tf.function` enables optimized, portable computational graphs.
 - Enterprise adoption: Strong integration with cloud platforms (GCP, Kubernetes).
 - Model serving: TensorFlow Serving for scalable, high-performance inference.
- TensorFlow excels in industry grade production deployment and scalability.





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What is PyTorch?

Pytorch is an open-source machine learning framework widely used for building and training deep neural networks. It is known for its flexibility, Pythonic design and use of dynamic computational graphs, which make it a popular choice for academic research and industry applications.

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PyTorch attributes

- The basic data structure for PyTorch are Tensors just like in TensorFlow.
- Automatic Differentiation (Autograd): Pytorch uses a tape-based differentiation engine called "Auto-grad", to calculate gradients automatically during the training process (backpropagation). This simplifies the implementation of complex neural networks and makes debugging easier.





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PyTorch key Features

- **Pytorch is designed to be deeply integrated with Python and compatible with Python libraries like NumPy and SciPy. This make it intuitive and easier to learn for developers who are already familiar with Python.**

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PyTorch ecosystem

- **PyTorch has a rich ecosystem of domain-specific libraries and tools such as torchvision (for computer vision), torchaudio (for audio processing) and torchtext (for Natural Language Processing tasks)**



PyTorch primarily supports Python, with available C++ frontend (LibTorch) and Java support for specific deployment needs. TensorFlow offers a broader range of language support including Python, C++, Java and JavaScript.

Feature	TensorFlow	PyTorch
Primary Strength	Production & deployment	Research & Experimentation
Learning Curve	Steeper	Easier, more intuitive
Computational Graph	Static +dynamic (tf.function)	Dynamic (eager by default)
High-level API	Keras (built-in)	Optional (Lightning, etc)
Visualization	TensorBoard (native)	TensorBoard (Third Party)
Mobile & Edge	TensorFlow Lite	Limited (Torch Mobile)
Model Serving	TensorFlow Serving	TorchServe
Ecosystem	Very large, end-to-end	Research-focused
Web Support	TensorFlow.js	Limited
Flexibility	Lower-level restrictions	More flexible

Conclusion

Both PyTorch and TensorFlow are fundamental for machine learning and deep learning projects, so they mostly serve the same purpose in software development and have very similar goals. Their difference lie in specific projects situations, user experience and preferences.